

Bitcoin Network Types & Hands-on Regtest

with bitcoin-cli



Why multiple networks?

Bitcoin doesn't operate on just one network. There are multiple environments designed for different purposes:

The three most important
Bitcoin networks are:



Real money



Mainnet



Testing



Testnet



Development



Regtest

Mainnet

Mainnet is the real Bitcoin network.



Characteristics:

- Real BTC with real value
- Full security
- Real miners
- Real economic activity



If you make a mistake here, you lose real money.

Testnet

Testnet is a public testing network.



Characteristics:

- Free coins (no real value)
- Anyone can join
- Same rules as Mainnet (mostly)



It's useful for:

- Wallet testing
- Application testing

Problems with Testnet



Limitations:

- Can be unstable
- Faucets unreliable
- Chain resets have happened

Regtest

Regtest stands for “Regression Test”.



You control everything. Regtest is:

- Local network (your machine)
- Instant block mining
- Fully customizable
- No external peers

Comparison Table

Feature	 Mainnet	 Testnet	 Regtest
Real money	Yes ✓	No ✗	No ✗
Public	Yes ✓	Yes ✓	No ✗
Mining	Real	Real-ish	Instant
Control	None	Low	Full